ISE-521 Sample Mid-Term

Enter student name: _____

Enter student USC ID number: _____

1) For each of the following scenarios, formulate the null and alternate hypotheses, identify the appropriate classical statistics hypothesis testing technique, specifying whether it is one-tailed or two-tailed, and whether it is one-sample, two-sample, or multi-sample:

1A) Researchers believe that college students at a university are getting less sleep than the recommended 8 hours per night. They collect a random sample of sleep durations from students at the university.

μ_0 = 8 hours

μ_1 = actual.

Null hypothesis:

getting more sleep $\mu_1 \geq \mu_0$.

Alternate hypothesis:

getting less sleep $\mu_1 < \mu_0$.

Classical statistics test:

One sample T- test

Number of tails:

1

Number of samples:

1

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1B) An e-commerce company has recently redesigned its checkout page and wants to test if the redesign has led to an improvement in the conversion rate (i.e., the proportion of visitors who make a purchase). They conduct an A/B test where half of the visitors are directed to the old checkout page and the other half to the new page.

Null hypothesis: P_{new} P_{old}.

new < old.

Alternate hypothesis:

new ≥ old.

Classical statistics test:

z-test

Number of tails:

1

Number of samples:

2.

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1C) An educational researcher hypothesizes that interactive teaching methods do not have the same effect on student test scores as traditional lecture-based methods. To test this, scores from classes using each method are collected.

Null hypothesis:

$\mu_0 = \mu_1$

μ_1 = interactive

μ_0 = traditional

Alternate hypothesis:

$\mu_0 \neq \mu_1$

Classical statistics test:

T-test

Number of tails:

2

Number of samples:

2

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1D) A university wants to examine if the major chosen by students (e.g., Engineering, Business, Arts) is independent of their gender. They collect data on the gender (male, female) and the major of several hundred students.

Null hypothesis:

.

dependent

Alternate hypothesis:

independent

Classical statistics test:

chi-square

Number of tails:

2

Number of samples:

multi

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1E) A business manager suspects that customer satisfaction levels for their product are different across three regions: North America, Europe, and Asia. Customer satisfaction surveys are collected and scored from each region.

Null hypothesis:

NA = E = A

Alternate hypothesis:

NA, E, A 其中一個 dif.

Classical statistics test:

one way ANOVA. (只有1e satisfaction level)

Number of tails:

multi

Number of samples:

multi

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2) A pharmaceutical company has developed a new hand sanitizer which it claims is more effective at reducing the presence of bacteria on hands compared to the currently leading brand. To validate this claim, an independent laboratory conducts a controlled experiment summarized below:

- Participants: 200 participants are randomly divided into two groups of 100 each.
- Group A (Test Group): Uses the new hand sanitizer.
- Group B (Control Group): Uses the leading brand hand sanitizer.
- Procedure: Each participant uses the hand sanitizer provided to them. Afterwards, their hands are swabbed and cultured to check for bacterial growth.
- Outcome Measurement: The outcome is binary, indicating the presence (1) or absence (0) of bacterial growth in the cultures.

Data collected:

- In Group A (new sanitizer), 25 out of 100 participants showed bacterial growth.
- In Group B (leading brand), 35 out of 100 participants showed bacterial growth.

2A) Formulate the hypotheses:

Null = Group A $\leq$ Group B

Alter = Group A > Group B.

2B) Calculate the test statistic (you do not need to do the full calculation, just give the formula with the appropriate numbers from the question statement above:

Two sample z-test

$$Z = \frac{P_1 - P_2}{\sqrt{P_{all}(1-P_{all})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{0.25 - 0.35}{\sqrt{0.3 \cdot 0.7\left(\frac{1}{100}+\frac{1}{100}\right)}} = -1.5.$$

$$P_{all} = \frac{25+35}{200} = 0.3.$$

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2C) Now you run a permutation test on these results with 100 permutations and the differences between ratios for the permutations are given below:

$P_1 - P_2 = -0.1$

-0.16, -0.14, -0.12, -0.12, -0.1, -0.08, -0.08, -0.08, -0.08, -0.06
-0.06, -0.06, -0.06, -0.06, -0.06, -0.06, -0.06, -0.04, -0.04, -0.04
-0.04, -0.04, -0.04, -0.04, -0.04, -0.04, -0.04, -0.04, -0.02, -0.02
-0.02, -0.02, -0.02, -0.02, -0.02, -0.02, -0.02, -0.02, -0.02, -0.02
-0.02, -0.02, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
0.0, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02, 0.02
0.02, 0.02, 0.02, 0.02, 0.04, 0.04, 0.04, 0.04, 0.04, 0.04
0.04, 0.04, 0.04, 0.04, 0.04, 0.04, 0.04, 0.04, 0.04, 0.04
0.04, 0.06, 0.06, 0.06, 0.06, 0.06, 0.06, 0.06, 0.06, 0.08
0.08, 0.08, 0.08, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.14

Based on these results, what would be the calculated p-value of the hypothesis test (use a one-tailed test)?

$\frac{4}{100} = 0.04$

H_0 = 没有差异
H_A = 有 "

如 $p < \alpha$

拒绝 H_0

A, B 分辨有差异

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3) Background: Consider a small dataset of the weights of pumpkins harvested from a local garden. The goal is to estimate the variability in the sample median weight using bootstrapping.

Dataset: Suppose the weights (in kilograms) are as follows: Weights= [2,3,5,7,9] $\bar{x}=5$

3A) What is the sample median?

5

3B) Below, manually construct five bootstraps:

2,2,2,2,2 随机放. 都 accept

3C) What is your estimate for the standard error of the median? You can leave your answer in the form of an equation.

都填上面数字 手机就不同